

DHRUV NEMANI

(408) 805-6716 | DhruvNemani@gmail.com | linkedin.com/in/dhruv-nemani | dnemni.github.io
Saratoga, CA 95070

EDUCATION

Purdue University

Aug. 2024 – May 2028

B.S. in Computer Science & Artificial Intelligence; Minor in Psychology

West Lafayette, IN

- **Relevant Courses:** Advanced Programming in C & Java, Data Structures & Algorithms, Computer Architecture, Advanced Probability & Statistics, Linear Algebra, Python Programming for Data Analytics, Systems Programming, Artificial Intelligence, Intro to Robotics (current), Data Mining & Machine Learning (current)
- **Honors/Awards:** Dean's List, Fall 2024 – Spring 2026; Semester Honors, Fall 2024 & Spring 2025

EXPERIENCE

Software Engineering Intern, Human-Machine Engineering Group

May 2026 – Present

Johns Hopkins University Applied Physics Laboratory

Laurel, MD

- Selected to continue part-time during the 2026–27 academic year after completing full-time summer internship
- **ORCA MCP:** Researched and developed a 31-tool MCP server, agent, and frontend enabling NLP/LLM-driven information retrieval and analysis of records with evidence-grounded answers via PostgreSQL database queries
- **Offline ORCA:** Added multi-wildcard FlexSearch, faceted filtering, click-to-scroll highlighting, and flat-file compatibility that is foundation for mission-critical offline document search for disconnected environments
- **Ticket Automation:** Prototyped a FastAPI/SvelteKit tool that ingests 4 note formats and extracts reviewed, citation-backed GitLab drafts with inline annotations; added Cypress test suite for troubleshooting and reliability
- **NVIT:** Added Cesium 3D map and offline MBTiles to Docker container, supporting 10,000+ tile files

Co-Founder & Platform Lead Engineer

Nov. 2025 – Present

CravingChef

Remote

- Cofounded an 8-person product team and lead backend/platform development for an Expo/React Native recipe app combining personalized discovery, meal planning, nutrition tracking, and grocery workflows
- Own Node.js APIs, content ingestion, analytics, and AI recommendations; added Redis caching for scalability with Firestore fallback and stamped protection to reduce repeated Firestore reads within 6-hour TTL window

Software Engineering Intern

May 2025 – Aug. 2025

Software Developers Inc.

Saratoga, CA (Remote)

- Developed POC agentic AI workflow for travel agent client to book flights and make itinerary
- Built AI tools for UI generation from image mockup to full MongoDB integrated project; delivered weekly demos

PROJECTS

RISC-V Virtual Machine | C17, RV32I, Software Systems, Linux/Unix

Jul. 2026 – Present

- Researched RISC-V emulation and created a 13-chapter learn-while-building plan that pairs systems concepts with incremental C17 implementation, progressing toward RV32I execution and eventual Linux boot

Applihero | Next.js, TypeScript, Supabase, PostgreSQL, OpenAI, RAG

Nov. 2025 – Present

- Shipped a full-stack AI career-coaching web application in 7 days and grew further with authentication, persistent data, vector embeddings, and semantic information retrieval over resumes and job descriptions with human review

CourseMate (POC) | Next.js, AWS S3, PostgreSQL, RAG

May 2025 – Aug 2025

- Built a full-stack course-management web application with PostgreSQL, calendar sync, progress tracking, RAG-based micro-learning plans, and generated study materials

LEADERSHIP

President & Founding Member

Aug. 2024 – Present

Purdue Squash Club

West Lafayette, IN

- Founded & manage a 100+ member club; captained team to 3rd & 2nd pl. in div at CSA Club Team Nationals

TECHNICAL SKILLS

Languages & Tools: Java, C, C++, Python, JavaScript, SQL, React, Expo, Vercel, Docker, Git, AWS, Linux, MCP

Data/Analysis: Quantitative Analysis, Data Collection, Modeling, Supabase, PostgreSQL, MongoDB, Redis, Firestore

Technical Domains: Natural Language Processing, Machine Learning, Information Retrieval, Mobile Application Development, Web Application Development, AI/ML, Software Engineering, Reliability Engineering, Automation